	Δ electricity [%]		Δ natural gas [%]		Δ discomfort [K·h]		
	summer	winter	summer	winter	summer	winter	
ZN-T drift (Core_mid, +0.5 K/d)	+1.6	-0.1	+10.2	-1.7	-1.2	+5.6	*sensor*
SA-T bias (AHU-2, +2 K)	+0.4	+7.1	+49.4	+5.5	-1.1	+0.8	
OA-T bias (+4 K)	-0.7	-0.4	0.0	+4.3	0.0	-0.2	
MA-T stuck (AHU-2)	0.0	+0.2	0.0	+21.1	0.0	0.0	
CHW-ST bias (-1.5 K)	-1.6	-0.4	0.0	0.0	0.0	0.0	
DA-F gain (VAV-Core_mid, +30%)	-0.2	+2.0	-0.1	-8.2	+0.1	0.0	
ZN-T dropout (Core_mid)	+4.4	-0.1	+87.3	-1.7	+167	+5.7	
VAV damper stuck 20% (Core_mid)	-0.6	+2.4	-1.8	-10.6	+4.0	0.0	*actuator*
Reheat valve stuck closed (Perim_mid_1)	0.0	0.0	0.0	-3.4	0.0	+162	
Reheat valve leaking (Perim_mid_1)	+2.5	+0.5	+1473	+31.0	+114	+71.3	
OA damper stuck 100% (AHU-2)	+3.0	+0.3	0.0	+157	0.0	+1.2	
OA damper stuck closed (AHU-2)	-1.8	+17.0	0.0	-15.4	0.0	-0.1	
CHW valve stuck closed (AHU-1)	+0.7	0.0	0.0	0.0	+1210	0.0	
Supply fan failure (AHU-1, 8 h)	0.0	0.0	0.0	-0.3	+219	+37.8	
Chiller fouling (CH-1, r=0.6)	+1.0	0.0	0.0	0.0	0.0	0.0	*equipment*
Chiller trip (CH-1)	-5.3	0.0	-0.9	0.0	+9.5	0.0	
Boiler fouling (r=0.6)	0.0	0.0	+7.7	+55.1	0.0	+1.6	
Tower fouling (r=0.5)	+3.3	0.0	0.0	0.0	0.0	0.0	
Filter fouling (AHU-2, +300 Pa)	+0.7	+0.8	0.0	-0.6	0.0	0.0	
Fan degradation (AHU-2, -30%)	+1.0	+1.1	0.0	-0.8	0.0	0.0	
Schedule 24/7 (AHU-2)	+14.4	+1.0	+149	+22.3	-0.3	+2.8	*control*
Economizer disabled (AHU-2)	-0.6	+6.4	0.0	-2.7	0.0	0.0	
PI mistuning x6 (VAV-Core_mid)	0.0	0.0	+0.7	+0.2	-0.1	+0.3	
Comm. loss VAV-Core_mid (12 h)	0.0	0.0	0.0	0.0	0.0	0.0	*network*
Occupancy surge x2 (Core_mid, 4 h)	0.0	0.0	0.0	-0.1	0.0	0.0	*event*
Heat wave / cold snap (+-5 K, 3 d)	+4.5	+0.1	-6.5	+22.5	+5.1	+2.0	
Window open (Perim_bot_1, 6 h)	0.0	0.0	0.0	+0.2	0.0	0.0	

colour: signed log scale; blue = decrease, red = increase vs. fault-free run